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A Careful Examination of Large Language Model Performance on Grade School Arithmetic

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Abstract

Large language models (LLMs) have achieved impressive success on many benchmarks for mathematical reasoning. However, there is growing concern that some of this performance actually reflects dataset contamination, where data closely resembling benchmark questions leaks into the training data, instead of true reasoning ability. To investigate this claim rigorously, we commission *Grade School Math 1000* (GSM1k). GSM1k is designed to mirror the style and complexity of the established GSM8k benchmark, the gold standard for measuring elementary mathematical reasoning. We ensure that the two benchmarks are comparable across important metrics such as human solve rates, number of steps in solution, answer magnitude, and more. When evaluating leading open- and closed-source LLMs on GSM1k, we observe accuracy drops of up to 8%, with several families of models showing evidence of systematic overfitting across almost all model sizes. Further analysis suggests a positive relationship (Spearman's $r^2 = 0.36$) between a model's probability of generating an example from GSM8k and its performance gap between GSM8k and GSM1k, suggesting that some models may have partially memorized GSM8k. Nevertheless, many models, especially those on the frontier, show minimal signs of overfitting, and all models broadly demonstrate generalization to novel math problems guaranteed to not be in their training data.

1 Introduction

Improving reasoning in large language models (LLMs) is one of the most important directions of current research. As such, proper benchmarking of current LLM abilities is paramount for ensuring progress continues in the correct direction. Currently, the field typically relies on public benchmarks such as GSM8k (Cobbe et al. [2021]), MATH (Hendrycks et al. [2021b]), MBPP (Austin et al. [2021]), HumanEval (Chen et al. [2021]), SWEBench (Jimenez et al. [2024])). However, because LLMs are trained on large corpora of data scraped from the Internet, there are major concerns that such benchmarks may inadvertently include examples that closely resemble the questions found in such benchmarks. This contamination may result in models having weaker reasoning capabilities than otherwise believed, due to simply being able to repeat the correct answer that it

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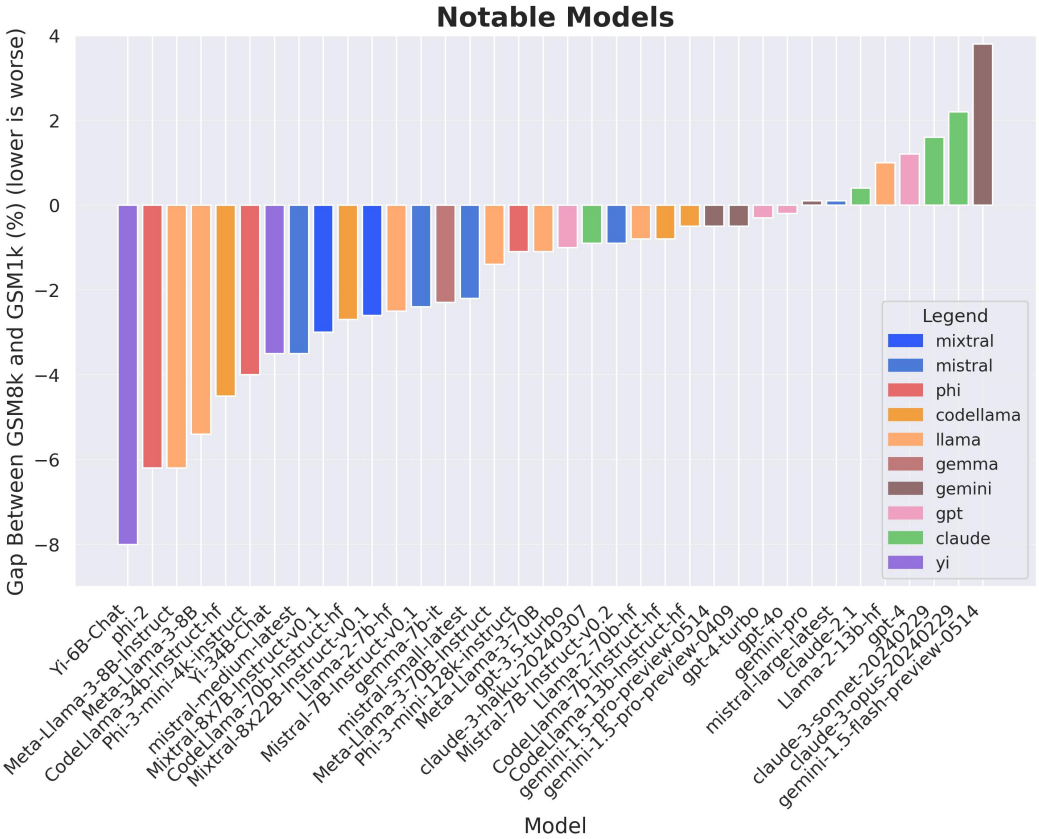

Figure 1: Notable models arranged by their drop in performance between GSM8k and GSM1k (lower is worse). We notice that Phi, Mistral and some models in the Llama family seem to be overfitting GSM8k, while models such as Gemini, GPT, and Claude show little to no signs of overfitting.

previously encountered during pre- or post- training. To properly investigate the reasoning abilities of models, we commission GSM1k, a newly constructed collection of 1205 grade school level math problems designed to mirror that of GSM8k. We take extensive efforts to ensure that GSM1k has a similar distribution of difficulty to GSM8k to ensure an apples-to-apples comparison. These efforts are described in Section 3, alongside a detailed description of the data creation process. To mitigate worries about data contamination, we created GSM1k solely with human annotators, without assistance from any LLM or other synthetic data source.

Dataset	**Example**
GSM8k	James writes a 3-page letter to 2 different friends twice a week. How many pages does he write a year?
GSM1k (ours)	Lee bought 6 shares of Delta stock at $40 per share. If he wants to make $24 from this trade, how much should Delta stock be per share when he sells?

Figure 2: Example from both the GSM8k dataset and the new GSM1k dataset (ours). We also provide an additional 50 examples from GSM1k in Appendix G.

We benchmark leading open- and closed-source LLMs on GSM1k, including GPT-4 (OpenAI et al. [2024]), Gemini (Team et al. [2024]), Claude, Mistral (Jiang et al. [2024, 2023]), Llama (Touvron et al. [2023a,b]), Phi (Gunasekar et al. [2023], Abdin et al. [2024]) and many more. Our analysis confirms the widespread suspicion in the field that many models are contaminated by benchmark data, with the worst models performing 8% worse on GSM1k compared to GSM8k. Additionally, our results

suggest that several families of models show consistent evidence of overfitting for nearly all model versions and sizes. Further analysis finds a positive relationship (Spearman's $r^2 = 0.36$) between a model's probability of generating examples from GSM8k and its performance gap between GSM8k and GSM1k, strongly suggesting that one important component of this overfitting is that models have partially memorized examples from GSM8k. Nevertheless, our results find that all frontier models show minimal signs of overfitting. Additionally, we also find that all models, including the most overfit ones, are still capable of successfully generalizing to new mathematical grade school problems, albeit occasionally at lower rates than their benchmark numbers would suggest.

We do not intend to release GSM1k publicly at this time to prevent a similar problem of data contamination occurring in the future. However, we plan to run recurring evaluations of all major open- and closed- source releases and to continually update our results. We will also open source our entire evaluation code so that the public version of our results can be reproduced. Additionally, we commit to open sourcing the entire benchmark when either 1) the top open source models score over 95% on GSM1k or 2) June 2025, whichever comes earlier. See Section 3 for precise release criteria.

2 Related Work

A major inspiration of this work was the celebrated study on overfitting done on ImageNet classifiers in 2019 (Recht et al. [2019]). This work measured overfitting in ImageNet by creating new versions of CIFAR10 and ImageNet and measuring the performance gap between the public test set and the newly created sets they constructed. In this work, we do a similar analysis on GSM8k, one of the leading benchmarks for mathematical reasoning. GSM1k is modelled after the GSM8k dataset (Cobbe et al. [2021]), released by OpenAI in 2021, which consists of 8.5k grade school math problems. Each problem is designed to be solvable using only basic arithmetic operations ($+$, $-$, $\times$, $\div$) with a difficulty level appropriate for grade school students. As of June 2024, top models report benchmark accuracies of over 95% (Team et al. [2024]). Other popular benchmarks for reasoning include MATH (Hendrycks et al. [2021b]) , MMLU (Hendrycks et al. [2021a]), GPQA (Rein et al. [2023]).

2.1 Data Contamination

Because data contamination is a well known issue in the field (Balloccu et al. [2024], Magar and Schwartz [2022], Sainz et al. [2023], Jacovi et al. [2023], Xu et al. [2024]), model builders will frequently take great pains to minimize the likelihood of data contamination. For example, it is common to remove all data with too high of an n-gram overlap with the benchmark data (Brown et al. [2020]). Additionally, methods such as using embedding similarity attempt to remove all contaminated data that is too similar in embedding space to the dataset (Shi et al. [2024]).

Xu et al. [2024] propose using similar variants of a benchmark questions to detect if models favor the original wording as a proxy for data contamination. Srivastava et al. [2024] propose functional evaluations, where benchmarks are written in the form of functions that can generate an infinite number of specific evaluation datapoints, each with slightly different numbers. In this setup, whenever a language model is evaluated, functional evaluations generate a specific problem instance to evaluate the model on, which is then never used again. This reduces the worry of data contamination by ensuring that no datapoint is ever used twice. Like ours, their results indicate the LLMs may be severely overfit on benchmark data. The main advantage of our approach over a purely function based evaluation is that functional evaluations can only generate a tiny portion of the full problem space by producing variations of the same problem with slightly different numerical values. Their results also suggest substantial amounts of data contamination, including for frontier models, in the MATH dataset.

3 GSM1k

GSM1k consists of 1205 problems requiring only elementary mathematical reasoning to solve. We created GSM1k using human annotators. Annotators were prompted with 3 example GSM8k problems and asked to produce novel problems of a similar difficulty level. The precise instructions and UI given to the annotators is available in Appendix C. All problem annotators were instructed to create problems solvable with only basic arithmetic (addition, subtraction, multiplication, and